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Scaling Laws for SAE Training Data

SAE Quality Depends on Buffer Diversity, Not Fresh Tokens: 8–64× Storage Reduction with Negligible Quality Loss

By Nikita Koriagin, Nikita Balagansky, Daniil Gavrilov (T-Tech)

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Training large sparse autoencoders is bottlenecked by activation storage. The authors show that SAE quality depends on the number of unique activations in the buffer—not on whether each training step uses fresh tokens. After a short diversity-limited regime, replay of a fixed buffer is as good as continuous fresh-data collection, cutting storage requirements by 8–64× with negligible quality loss across two model families, three layers, three SAE architectures, and four evaluation metrics.

AT A GLANCE

Problem: SAE training requires storing billions of LLM activations; the activation buffer dominates I/O cost and disk usage.

Why it matters: SAEs are a leading interpretability tool; reducing their training cost unlocks larger-scale and more accessible interpretability research.

Method: Fit a data-constrained scaling law; identify the unique-activation threshold n*; collect n* diverse activations once, then replay.

Result: 8–64× reduction in activation storage; <1% quality loss on reconstruction and interpretability metrics.

Evidence: Llama-3.1-8B and Qwen3-7B; early/middle/late layers; BatchTopK, TopK, JumpReLU SAEs; 50M–10B token budgets.

The Big Picture

mechanistic interpretability: trained on the residual stream activations of a frozen LLM, they decompose those activations into sparse linear combinations of learned features, each of which often corresponds to an interpretable concept.

The standard recipe—generate fresh activations for every training step from a streaming corpus—implicitly assumes that data novelty is the scarce resource. The central finding of this paper is that this assumption is wrong for SAEs: the scarce resource is *activation diversity* (coverage of the activation distribution), not novelty (seeing each activation only once). Once the buffer achieves adequate coverage, additional unique activations provide negligible benefit, and the buffer can be replayed indefinitely.

A single residual stream layer of Llama-3.1-8B at fp16 precision generates 16 KB per token, so 1B tokens require 16 TB of storage. This activation bottleneck is a major practical obstacle to large-scale SAE research, particularly for researchers without datacenter-scale storage infrastructure.

Why Existing Approaches Fall Short

Streaming activation generation (the standard recipe) generates fresh activations for every training step. It is correct but expensive: it ties training speed to LLM inference throughput and requires either online inference or offline storage at full token budget.

Fixed small buffers without principled sizing have been used in practice (e.g., the OpenAI SAE release uses a buffer of 400K activations) but without a principled criterion for the right buffer size. Too small a buffer leads to mode collapse; too large is unnecessarily costly.

This paper is the first to characterize the data-efficiency of SAE training formally, through the lens of a data-constrained scaling law, and to derive the optimal buffer-sizing strategy from first principles.

Core Mechanism

numbers of *unique* activations but different levels of repetition (one streaming, one with replay) produce SAEs of comparable quality. This decouples the training budget (number of gradient steps) from the data budget (number of unique activations).

The unique-activation count n_{unique} is the number of distinct activation vectors that the SAE has seen at least once during training. The authors find empirically that SAE quality $\mathcal{L}$ is well described by a power law:

$$\mathcal{L}(n_{unique}) = \mathcal{L}_\infty + C \cdot n_{unique}^{-\alpha}$$

where $\mathcal{L}_\infty$ is the asymptotic quality, $C > 0$ is a scaling constant, and $\alpha > 0$ is the data-scaling exponent. This power law holds across model families, layer positions, SAE architectures, and evaluation metrics.

Technical Formulation

SAE training minimizes:

$$\mathcal{L}_{SAE}(W_e, W_d, b) = E_x[||x - W_d\ \mathrm{TopK}(W_e x + b)||_2^2 + \lambda||\mathrm{TopK}(W_e x + b)||_1]$$

The **data-constrained scaling law** is fit by: (1) train multiple SAEs with the same gradient-step budget T but varying buffer sizes; (2) measure final quality $\mathcal{L}_i$ for each run; (3) fit the three parameters $(\mathcal{L}_\infty, C, \alpha)$ via non-linear least squares. The optimal buffer size n^* is defined as the smallest n_{unique} such that $\mathcal{L}(n^*) \le \mathcal{L}_\infty + \delta$ for a chosen tolerance δ (e.g., 1% above $\mathcal{L}_\infty$).

Learning or Inference Procedure

The practical buffer-sizing workflow: (1) Run a small calibration sweep: train 5–10 SAEs with varying buffer sizes at 1% of the full training budget; (2) Fit the scaling law to the calibration sweep results; (3) Identify n^* from the fitted law; (4) Collect n^* activations from the LLM (a one-time cost); store this buffer on disk; (5) Train the full SAE by replaying activations sampled uniformly from the buffer. The calibration sweep adds 10% overhead relative to a single training run; this is recovered in full by the reduced storage and I/O cost.

What the Guarantee Says

The empirical claim is: across all evaluated settings (Llama-3.1-8B, Qwen3-7B; layers 4, 16, 28; BatchTopK/TopK/JumpReLU; reconstruction loss, dead-neuron fraction, feature absorption, automated interpretability), training with the optimal-buffer-size replay strategy degrades quality by less than 1% relative to the streaming baseline at a storage reduction of 8–64×.

Experimental Findings

Setting	Storage reduction	Quality loss
Llama-3.1 layer 4	64×	<0.5%
Llama-3.1 layer 16	32×	<0.3%
Llama-3.1 layer 28	16×	<0.4%
Qwen3 all layers	8–32×	<1%
Automated interp.	16×	<2%

The largest storage reductions occur in early layers, where activations are more stereotyped and coverage is achieved with fewer unique examples. Late layers, where activations carry more semantic diversity, require more unique examples but still admit substantial savings.

Ablations and Interpretation

Scaling law fit quality: The power-law form fits observed data with $R^2 > 0.98$ in all 18 evaluated settings (3 layers × 3 architectures × 2 models), confirming the law is not an artifact of a particular setup.

Scaling exponent α: Values range from 0.3 (early layers, Llama-3.1) to 0.7 (late layers, Qwen3). Higher α means quality improves more steeply with n_{unique}; these settings benefit less from replay.

Uniform vs. prioritized replay: Replaying activations proportional to their reconstruction loss improves quality by <0.2% over uniform replay; uniform replay is recommended for simplicity.

Generalization to new domains: A SAE trained with a Wikipedia-only buffer and deployed on code activations degrades by 8–12%, confirming that buffer diversity must match the deployment distribution.

Reference

Nikita Koriagin, Nikita Balagansky, Daniil Gavrilov. **Scaling Laws for SAE Training Data.** NeurIPS 2026 Poster #148634. neurips.cc/virtual/2026/poster/148634